THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which triangular number comes just after 190? 1

(a) 229 **(b)** 211 **(c)** 210 **(d)** 209

2. A reflex angle is $200°$. The angle needed to complete a full turn is: 1

(a) 150 **(b)** 170 **(c)** 160 **(d)** 250

3. Starting from 15, the Collatz sequence reaches 1 in how many steps? 1

(a) 16 **(b)** 18 **(c)** 19 **(d)** 17

4. What fraction of the total do Tennis players form? 1

Sport	Students
Tennis	35
Hockey	60
Football	50
Cricket	60

(a) $\frac{8}{41}$ **(b)** $\frac{12}{41}$ **(c)** $\frac{7}{41}$ **(d)** $\frac{1}{6}$

5. The smallest digit to put in the blank to make 452_ divisible by 3 is: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** 4

6. A pattern of stacked sticks uses 3 sticks for 1 triangle and 2 more sticks for each extra triangle added in a row. Sticks for 10 triangles: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 20

7. A ray OB bisects $\angle AOC = 54°$. Then $\angle AOB$ is: 1

(a) 27 **(b)** 32 **(c)** 37 **(d)** 22

8. Round 48376 to the nearest thousand, and the result is 48000. The difference between 48376 and the rounded number is: 1

(a) 375 **(b)** 386 **(c)** 376 **(d)** 377

9. Items sold on 3 days: 25, 35, 30. The average per day is: 1

(a) 30 **(b)** 31 **(c)** 29 **(d)** 35

10. The largest 3-digit multiple of 7 is: 1

(a) 994 **(b)** 995 **(c)** 1001 **(d)** 987

11. The sum of the first 24 natural numbers added to the sum of the first 25 natural numbers is: 1

(a) 626 **(b)** 649 **(c)** 624 **(d)** 625

12. 4 points lie on a line. The number of rays that start at one of the points and pass through another point is: 1

(a) 13 **(b)** 12 **(c)** 11 **(d)** 16

13. The sum of the first 30 counting numbers 1, 2, 3, 4, ... is: 1

(a) 466 **(b)** 495 **(c)** 465 **(d)** 435

14. The data 2, 4, 1, 4, 3, 5, 2, 3, 5, 4, 6, 6, 5, 1, 1, 3, 6, 1, 3, 6, 2, 4, 4, 6 has the value 6 how many times? 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 7

15. How many prime numbers are there between 20 and 60? 1

(a) 11 **(b)** 8 **(c)** 10 **(d)** 9

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Is 36 a term of $3, 7, 11, 15, \ldots$? Give a reason. 2

17. OB bisects $\angle AOC$ and $\angle AOB = 28°$. Find $\angle AOC$ and its type. 2

18. Apply the reverse-and-add steps on 87 and state the palindrome reached. 2

19. In a pictograph, one symbol = 10 students. 5 symbols show Class VI-A. Class VI-B has 20 students more. How many symbols show VI-B? 2

20. Find the smallest number that must be added to 1234 to make it divisible by 9. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the largest palindrome less than 1300 and the smallest palindrome greater than 1300. 3

22. A shop sold 40, 60, 80 and 100 kg of rice in four weeks. Find the total, and draw a bar graph scale: with 1 cm = 20 kg, find the heights. 3

23. Two bells ring every 12 minutes and every 18 minutes. They ring together at 9 am. When do they next ring together? 3

24. Write the 20th and 21th triangular numbers and check that they add to a square. 3

OR

There are 5 points on a line. How many line segments can be named? How many of them do not include the end point A? 3

25. Find the sum of all numbers from 1 to 40 by pairing the first and last numbers. 3

OR

In a class of 20 students the data of the number of books read in a month is 3, 0, 1, 0, 4, 5, 4, 2, 1, 5, 2, 3, 3, 3, 1, 1, 1, 5, 2, 3. How many read more than 2, and how many read exactly 2? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A room is 450 cm long and 360 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Show that the sum of the cubes of the first 9 numbers is a perfect square, and find the sum of the first 9 numbers. 5

27. Four rays OA, OB, OC, OD are drawn from O such that $\angle AOB = 40°$, $\angle BOC = 55°$ and $\angle COD = 70°$, measured one after the other. Find $\angle AOD$ and the angle needed to complete a full turn. 5

OR

Using the digits 3, 7, 5, 1 once each, form a 4-digit number closest to 5000. Show the options you tried and the difference. 5

END OF QUESTION PAPER **Total: 50 Marks**

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